

Accenture Assessment — Consolidated Practice Question Bank

Compiled from 8 past-paper PDFs: Actual Paper MCQs, Coding, Game-Based, SQL, Common Applications, Computer Science, Fundamentals of Networking, Pseudo Code.

~200 unique questions, de-duplicated and re-ordered into the sequence you should actually practise them in.

How to use this book

The original PDFs are organised by *where the question came from*. This one is organised by *what you need to learn first*. Each stage assumes the one before it.

Stage	Section	Qs	Suggested time
1	Pseudocode & Programming Logic	14	Day 1-2
2	SQL — Concepts (MCQ)	8	Day 2
3	SQL — Query Writing	30	Day 3-4
4	Coding Round — Tier 1 (Warm-up)	24	Day 1-5 (parallel)
5	Coding Round — Tier 2 (Core)	26	Day 5-9 (parallel)
6	Coding Round — Tier 3 (Hard)	18	Day 9-12 (parallel)
7	OOP & Programming Concepts	7	Day 6
8	Operating Systems & Computer Architecture	13	Day 7
9	Web Technologies & APIs	14	Day 8
10	DBMS Concepts	4	Day 9
11	Networking & Security	26	Day 10-11
12	Cloud & DevOps	27	Day 12-13
13	MS Office / Common Applications	16	Day 14
14	Game-Based / Logical Round	15	Day 15

Rules for practising

1. **Cover the answer.** Every answer sits directly under its question in a grey box. Fold the page or cover it with your hand. Reading the answer first feels like learning; it isn't.
2. **Run the coding stages in parallel from Day 1.** 2-3 problems a day, every day. Do not save them for the end — coding is the only section you cannot cram.
3. **Hand-trace every pseudocode question.** Write the variable values line by line on paper. The exam gives you no debugger.
4. Questions marked **[HIGH-FREQ]** appeared **two or more times** across the eight source papers. If you are short on time, do only these.
5. Section 15 is the **errata list** — places where the source PDFs give a wrong or contradictory answer. Read it before you memorise anything.

Stage 1 — Pseudocode & Programming Logic

Source: Pseudo Code.pdf. The single highest-yield section — pure output-prediction, no syntax knowledge needed, and you can be 100% accurate with practice.

Before you start, memorise these four operators:

Op	Name	Rule
&	bitwise AND	1 only when both bits are 1
|	bitwise OR	1 when either bit is 1
^	bitwise XOR	1 when bits differ
~	bitwise NOT	flips every bit

Q1.1 What will be the output of the following pseudo code?

```
Integer p, q, r
Set p=1, q=8, r=9
q = (11+r) & q
p = q + q
Print p+q+r
```

(a) 6 (b) 9 (c) 13 (d) 24

Answer: (b) 9. $q = (11+9) \& 8 = 20 \& 8$. $20 = 10100$, $8 = 01000$ — no shared set bits, so $q = 0$. Then $p = 0+0 = 0$. Output $0+0+9 = 9$.

Q1.2 What is the final value of `total` ?

```
FUNCTION incrementByThree(n)
    RETURN n + 3
END FUNCTION
total = 0
FOR i = 1 TO 3
    total = total + incrementByThree(i)
END FOR
DISPLAY total
```

(a) 15 (b) 12 (c) 18 (d) 9

Answer: (a) 15. The loop produces $1+3=4$, $2+3=5$, $3+3=6$. Total = $4+5+6 = 15$.

Q1.3 What does this pseudocode accomplish?

```
FUNCTION cal(number):
    IF number is 0 OR number is 1
        RETURN 1
    ELSE:
        RETURN some_operation(number, cal(number - 1))
    End IF
END FUNCTION cal
result = cal(input_number)
PRINT result
```

(a) Factorial of input_number (b) Sum of input_number and its predecessors (c) Product of predecessors (d) Square of input_number

Answer: (a) Factorial. Base case returns 1 at 0 or 1; each call multiplies n by the result for $n-1$. That is $n! = n \times (n-1) \times \dots \times 1$.

Q1.4 Output for a=2, b=8, c=8?

```

Integer funn(Integer a, Integer b, Integer c)
    c = a + a
    if((4+10*a) < (7+c))
        if((7+a) > c)
            if((5*b) < (b+5))
                b = a + b
            End if
        End if
    End if
    Print a+b+c

```

(a) 17 (b) 22 (c) 21 (d) 29

Answer: (c) 21. c becomes 11 (per the source's trace: $c = 9 + a$). Outer test $(4+20) < (7+11) \rightarrow 24 < 18$ is **false**, so nothing inside runs and b stays 8. Output = $2+8+11 = 21$. *Note the trap: the answer depends entirely on the first condition being false.*

Q1.5 Output of the following 2-D array pseudo code?

```

Integer j
Integer arr[2][2] = {{0, 1}, {4, 1}}
if((3 & arr[1][1]) < (arr[1][0] + 3))
    arr[1][0] = (4 + 1) + arr[0][1]
else
    arr[1][0] = 2 + arr[1][0]
arr[0][0] = (arr[0][0] ^ 1) + arr[0][0]
Print arr[1][0] + arr[0][0]

```

(a) 7 (b) 12 (c) 9 (d) 19

Answer: (a) 7. $3 \& 1 = 1$, $4+3 = 7$, so $1 < 7$ is true $\rightarrow arr[1][0] = 5+1 = 6$. Then $arr[0][0] = (0^1)+0 = 1$. Output = $6+1 = 7$.

Q1.6 Output of the given pseudo code?

```

Integer j
Integer arr[2][2] = {{1, 2}, {2, 4}}
arr[0][1] = (arr[1][0] + arr[0][0]) & arr[1][1]
if((arr[0][0] - arr[1][1]) > (arr[1][1] + arr[0][0]))
    arr[0][0] = (arr[1][1] + arr[1][1]) + arr[0][0]
End if
arr[0][0] = (3 + 7) & arr[0][0]
Print arr[1][1] + arr[0][0]

```

(a) 1.0 (b) 4.0 (c) 8.0 (d) 17.0

Answer: (b) 4.0. $(2+1) \& 4 = 3 \& 4 = 0$. The if is false. $arr[0][0] = 10 \& 1 = 0$. Output = $4 + 0 = 4$.

Q1.7 Output for a=7, b=4, c=10?

```

Integer funn(Integer a, Integer b, Integer c)
    if((b+c+a) < (c+a+b))
        b = b + c
    Else
        if(3+c < (a+3))
            a = (4+5) + b
        End if
        c = c + a
    End if
    c = (b+b) + a
    Print a+b+c

```

(a) 24 (b) 30 (c) 26 (d) 32

Answer: (c) 26. $21 < 21$ is false $\rightarrow$ else branch. $13 < 10$ is false. $c = 10+7 = 17$. Then $c = (4+4)+7 = 15$. Output = $7+4+15 = 26$. **Trap:** the final $c = (b+b)+a$ overwrites the 17.

Q1.8 Output for $a=1, b=2, c=10$?

```
Integer funn(Integer a, Integer b, Integer c)
    if((a+b)+(b+c) > (a+b+c))
        if(a > b)
            b = b + c
            if((c+a+b) < (a+b+c))
                c = (b+c) + c
            End if
        End if
    End if
    Print a+b+c
```

(a) 9 (b) 15 (c) 13 (d) 19

Answer: (c) 13. Outer: $(1+2)+(2+10) = 15 > 13$, true. Inner: $1 > 2$ is false — everything inside is skipped. Values unchanged: $1+2+10 = 13$. (The source PDF prints "26" as the answer here; its own trace says 13. See the errata at the end.)

Q1.9 Write pseudocode to initialise an array `ages` with `[25, 30, 22, 28, 35]` and print the **second** element.

- (a) `ARRAY ages[5] = [25,30,22,28,35] / PRINT ages[2]`
(b) five separate assignments then `PRINT ages[2]`
(c) `ARRAY ages[5] = [25,30,22,28,35] / PRINT ages[1]`

Answer: (c). Arrays are 0-indexed: `ages[0]=25`, `ages[1]=30`. The second element is at index 1. **[HIGH-FREQ]** — off-by-one indexing is tested repeatedly.

Q1.10 Which value of `s` prints "Valid"?

```
function validCheck(string s):
    if s matches "[a-zA-Z0-9]{6}"
        then print "Valid"
    else
        then print "Not Valid"
```

(a) `arun32` (b) `kkvarun32` (c) `arun$2` (d) `arun@2`

Answer: (a) arun32. `{6}` demands exactly six characters, `[a-zA-Z0-9]` allows only letters and digits. (b) has 9 characters; (c) and (d) contain `$` and `@`.

Q1.11 Output of this loop?

```
Integer a, b, c
Set a=8, b=5, c=6
for(each c from 3 to 4)
    b = 9 + b
    b = (b+7) + a
    b = (b&4) + b
End for
Print a+b
```

(a) 78 (b) 69 (c) 55 (d) 65

Answer: (d) 65. Iteration 1: $b=14 \rightarrow 29 \rightarrow (29\&4)+29 = 4+29 = 33$. Iteration 2: $b=42 \rightarrow 57 \rightarrow (57\&4)+57 = 0+57 = 57$. Output = $8+57 = 65$. **[HIGH-FREQ]** — loops combined with bitwise AND are the signature Accenture pseudocode question.

Q1.12 Output for $a=0, b=2, c=7$?

```

Integer funn(Integer a, Integer b, Integer c)
    if((a+c+b) < (3+b-a))
        a = b + c
        if((b+c+a) < (9+a+6))
            a = (b+a) + b
        Else
            b = (3+4) + a
        End if
        c = (b+c) + c
    End if
    Print a+b+c

```

(a) 6 (b) 9 (c) 13 (d) 27

Answer: (b) 9. $(0+7+2)=9 < (3+2-0)=5$? No. Entire block skipped. Output = $0+2+7 = 9$.

Q1.13 Output of the following?

```

Integer p, q, r
Set p=4, q=8, r=6
if((p+r) > (q-p))
    q = (q+7) + q
    if((4 + q^p) < (7+p+r))
        p = (q^p) ^ r
    Else
        r = 5 + q
    End if
End if
Print p+q+r

```

(a) 50 (b) 59 (c) 55 (d) 63

Answer: (c) 55. $10 > 4$ true $\rightarrow q = 15+8 = 23$. $23^4 = 19$, so $4+19 = 23 < 17$? False $\rightarrow r = 5+23 = 28$. Output = $4+23+28 = 55$.

Q1.14 Output for a=1, b=6, c=7?

```

Integer funn(Integer a, Integer b, Integer c)
    if((a+8+c) < (6+a))
        a = (a+b) + b
        if((a+b+c) < (b+c+a))
            a = c + c
        Else
            c = (c+a) + b
        End if
        c = 7 + b
    End if
    return a+b+c

```

(a) 14 (b) 20 (c) 9 (d) 24

Answer: (a) 14. $(1+8+7)=16 < (6+1)=7$? No. Nothing executes. Return $1+6+7 = 14$.

Stage 2 — SQL Concepts (MCQ)

Source: *Pseudo Code.pdf* (SQL MCQs) + *Computer Science.pdf*. Do these before writing queries.

Q2.1 What is the output of `SELECT emp_name FROM employees WHERE emp_name LIKE 'a%';`

- (a) Names starting with 'a' (b) Names ending with 'a' (c) Names having 'a' in any position (d) Names having 'a' in between

Answer: (c). `%` matches any sequence of characters (including none) on both sides, so 'a' may sit anywhere. Learn the whole family: `'a%'` = starts with, `'%a'` = ends with, `'_a%'` = 'a' as the 2nd character.

Q2.2 Which statement retrieves items in ascending order?

- (a) `SELECT * FROM [table] ORDER BY [tbl_clm];` (b) `... SORT BY ...` (c) `SELECT * FROM [table] ORDER BY [tbl_clm] ASC;` (d) `... GROUP BY [tbl_clm] ASC;`

Answer: (c). `ORDER BY` sorts; `ASC` states ascending explicitly. `SORT BY` is not standard SQL, and `GROUP BY` aggregates rather than sorts. (Note (a) is also functionally ascending since `ASC` is the default — the exam wants the explicit form.)

Q2.3 You must update the status of orders placed **before** 1 Jan 2021 to 'Shipped'. Which query is correct?

- (a) `UPDATE orders SET status='Delivered' WHERE order_date < '2021-01-01';`
(b) `UPDATE orders SET status='Shipped' WHERE order_date > '2021-01-01';`
(c) `UPDATE orders SET status='Shipped' WHERE status='2021-01-01';`
(d) `UPDATE orders SET status='Shipped' WHERE order_date < '2021-01-01';`

Answer: (d). `SET` assigns the new value; `WHERE` selects the rows. Read both halves — three distractors change exactly one token.

Q2.4 Which shows correct usage of the `GROUP BY` clause?

- (a) `SELECT column_name1, column_name2 FROM table WHERE column_name1 = value GROUP BY column_name1`
(b) `SELECT * FROM table WHERE column_name = value GROUP BY column_name`
(c) `SELECT column_name, aggregate_function(column_name) FROM table WHERE column_name operator value GROUP BY column_name`
(d) `SELECT * FROM table GROUP BY column_name`

Answer: (c). `GROUP BY` is used **with an aggregate function** (`COUNT`, `SUM`, `AVG`, `MAX`, `MIN`). `WHERE` filters rows *before* grouping; `HAVING` filters *after*.

Q2.5 Count occurrences of the character 'm' in `FirstandLastName` from `StudentDetails`. Which query is correct?

- (a) `FROM StudentDetails SELECT ... COUNT(...) - COUNT(REPLACE(...))`
(b) `SELECT FirstandLastName, COUNT(FirstandLastName) - COUNT(REPLACE(FirstandLastName,'m','')) FROM StudentDetails;`
(c) `FROM StudentDetails SELECT ... LENGTH(...) - LENGTH(REPLACE(...))`
(d) `SELECT FirstandLastName, LENGTH(FirstandLastName) - LENGTH(REPLACE(FirstandLastName,'m','')) FROM StudentDetails;`

Answer: (d). Two things are being tested at once: (1) `SELECT` must precede `FROM`, killing (a) and (c); (2) counting *characters* needs `LENGTH`, not `COUNT` — `COUNT` counts rows.

Q2.6 You want student names and marks, and students must still appear even when the marks record is absent. Which join?

- (a) a malformed `LEFT JOIN ... GROUP BY ... IN` (b) another malformed variant (c) `SELECT E.StudentName, M.Mark FROM StudentDetails E LEFT JOIN StudentMark M ON E.StudentId = M.StudentId;`

Answer: (c). `LEFT JOIN` keeps every row of the left table and fills `NULL` where the right table has no match. The join condition uses **ON**, never `IN` or `GROUP BY`.

Q2.7 A retail company's query `SELECT * FROM Customers WHERE Email = 'abc@example.com';` runs slowly on a 5-million-row table. Which optimisation helps most?

- (a) Adding more rows (b) Creating an index on the Email column (c) Using a foreign key on Email (d) Denormalising the table

Answer: (b) Index on Email. An index turns a full table scan into a lookup. This is the #1 database performance answer in campus papers.

Q2.8 A trigger must log every employee deletion into an `AuditLog`. Which structure is correct?

Answer: `AFTER DELETE ON Employees FOR EACH ROW INSERT INTO AuditLog (EmpID, DeletedOn) VALUES (OLD.EmpID, NOW());` — the log entry must be written *after* the delete, and deleted-row values are reached through the **OLD** keyword (`NEW` is for INSERT/UPDATE).

Stage 3 — SQL Query Writing

Source: Accenture SQL.pdf. In the real test you get a schema diagram and a blank editor. Practise writing these out by hand — do not just read them.

The pattern behind all 30 questions is identical:

1. Read the required output columns and their **aliases** — the grader checks column names.
2. Find which tables hold those columns in the schema diagram.
3. Chain JOINS along the foreign keys between them.
4. Translate the English filter into WHERE.
5. Add ORDER BY / GROUP BY / DISTINCT only if the question says so.

Q3.1 Transaction ID, amount and type of all transactions where type is "Debit" and amount is > 10000 but < 50000. Output:

TRANSACTION_ID | AMOUNT | TRANSACTION_TYPE

```
sql SELECT Transaction_ID, Amount, Transaction_Type FROM transaction WHERE Transaction_Type = 'Debit' AND Amount > 10000 AND Amount < 50000;
```

Q3.2 First name, contact number and balance of all customers whose account type starts with "Sa", ordered by first name.

```
sql SELECT c.First_Name, c.Contact, a.Balance FROM customer c JOIN account a ON c.Customer_ID = a.Customer_ID JOIN account_type at ON a.Account_Type_ID = at.Account_Type_ID WHERE at.Account_Type_Name LIKE 'Sa%' ORDER BY c.First_Name;
```

Three-table chain + prefix LIKE + ORDER BY. The single most representative question in the set.

Q3.3 Employee ID, name, basic salary and net pay for employees whose basic salary is greater than 5,000. Output: EMPID | EMPNAME | BASIC | NETPAY

```
sql SELECT emp_info.empid AS EMPID, emp_info.empname AS EMPNAME, salary_info.basic AS BASIC, emp_payroll.netpay AS NETPAY FROM emp_info JOIN salary_info ON emp_info.employee_category = salary_info.employee_category JOIN emp_payroll ON emp_info.empid = emp_payroll.empid WHERE salary_info.basic > 5000;
```

Q3.4 Names of employees with more than 5 years of experience who joined after 1 Jan 2001. Aliases "Employee ID", "Employee Name".

```
sql SELECT EMPID AS "Employee ID", EMPNAME AS "Employee Name" FROM employee_info WHERE YRS_OF_EXP > 5 AND JOINING_DT > '2001-01-01';
```

Q3.5 Course ID, course name, day and start time of all courses taught on 'Wednesday' (stored as 'wed').

```
sql SELECT c.course_id AS "Course ID", c.name AS "Course Name", s.day AS "Day", s.starttime AS "Start Time" FROM course c JOIN section sec ON c.course_id = sec.course_id JOIN schedule s ON sec.schedule_id = s.schedule_id WHERE s.day = 'wed';
```

Q3.6 Title, Price and ISBN of books published after 1 Jan 1940 in the "C102" category.

```
sql SELECT Title, Price, ISBN FROM books WHERE PublishedDate > '1940-01-01' AND Category = 'C102';
```

Q3.7 Category id and category name where the name starts with 'M'.

```
sql SELECT c.categoryid AS CATEGORYID, c.categoryname AS CATEGORYNAME FROM channelcategory c WHERE c.categoryname LIKE 'M%';
```

Q3.8 Train ID and name of all trains whose name starts with 'M' and which go to the station named "PUNE".

```
sql SELECT td.train_id, td.train_name FROM train_details_tbl td JOIN train_stations_tbl ts ON td.train_to = ts.station_id WHERE td.train_name LIKE 'M%' AND ts.station_name = 'PUNE';
```

Q3.9 Employee ID, leave type and total leaves for employees who took more than 10 leaves, where leave type is Casual (CL) or Medical (ML).

```
sql SELECT EMPID, LEAVE_TYPE, TOTAL_LEAVES FROM LEAVE_INFO WHERE TOTAL_LEAVES > 10 AND LEAVE_TYPE IN ('CL', 'ML');
```

Q3.10 ID, name, department name and basic salary of employees working in the 'HR' department.

```
sql SELECT ei.EMPID, ei.EMPNAME, di.DEPTNAME, si.BASIC FROM employee_info ei JOIN department_info di ON ei.DEPTID = di.DEPTID JOIN salary_info si ON ei.EMPLOYEE_CATEGORY = si.EMPLOYEE_CATEGORY WHERE di.DEPTNAME = 'HR';
```

Q3.11 Employee ID, name, department name and house rent allowance for employees in departments located in BANGALORE or COCHIN.

```
sql SELECT ei.EMPID, ei.EMPNAME, di.DEPTNAME, si.HOUSE_RENT_ALLOWANCE FROM employee_info ei JOIN department_info di ON ei.DEPTID = di.DEPTID JOIN salary_info si ON ei.EMPLOYEE_CATEGORY = si.EMPLOYEE_CATEGORY WHERE di.LOCATION IN ('BANGALORE', 'COCHIN');
```

Q3.12 Account type id and average account balance for each account type where the average balance is ≥ 50000 .

```
sql SELECT Account_Type_ID, AVG(Account_Balance) AS Average FROM Accounts GROUP BY Account_Type_ID HAVING AVG(Account_Balance) >= 50000;
```

The one GROUP BY + HAVING question in the set — learn it cold. HAVING filters groups; WHERE cannot contain an aggregate.

Q3.13 First name, last name and account id of customers with balance ≥ 50000 , ordered by first name.

```
sql SELECT First_Name AS FIRST_NAME, Last_Name AS LAST_NAME, Account_ID AS ACCOUNT_ID FROM customer c JOIN account a ON c.Customer_ID = a.Customer_ID WHERE a.Balance >= 50000 ORDER BY First_Name;
```

Q3.14 First name (alias "Staff First Name"), position and salary of staff where salary > 50000 .

```
sql SELECT firstname AS "STAFF FIRST NAME", position AS "POSITION", salary AS "SALARY" FROM staff WHERE salary > 50000;
```

Q3.15 Full name (first + last, alias PatientName), email, admission date and total billing for each patient with **unpaid** bills, sorted by total billing descending.

```
sql SELECT CONCAT(p.FirstName, ' ', p.LastName) AS PatientName, p.Email AS PatientEmail, p.AdmissionDate AS AdmissionDate, b.TotalAmount AS TotalBilling FROM Patient p JOIN Billing b ON p.PatientID = b.PatientID WHERE b.PaymentStatus = 'Unpaid' ORDER BY b.TotalAmount DESC;
```

[HIGH-FREQ] CONCAT for a full name appears three times across the papers — memorise `CONCAT(first, ' ', last)` with the space literal.

Q3.16 Flights operated by 'Singapore Airlines' — flight ID, departure date, departure time.

```
sql SELECT f.FLIGHT_ID, f.DEPARTURE_DATE, f.DEPARTURE_TIME FROM Flight f JOIN Airplane a ON f.AIRPLANE_ID = a.AIRPLANE_ID JOIN Airline al ON a.AIRLINE_ID = al.AIRLINE_ID WHERE al.NAME = 'Singapore Airlines';
```

Q3.17 Airplane ID and model number for airplanes manufactured by Airbus.

```
sql SELECT AIRPLANE_ID, MODELNUMBER FROM Airplane WHERE MANUFACTURER = 'Airbus';
```

Q3.18 Last names of students registered in the year 2012.

```
sql SELECT s.LAST_NAME FROM student s JOIN registration r ON s.STUDENT_ID = r.STUDENT_ID WHERE EXTRACT(YEAR FROM r.REG_DATE) = 2012;
```

Alternate: `WHERE r.REG_DATE LIKE '2012%'`. Know both — some graders reject one.

Q3.19 Cabin crew whose first name starts with 'A' and who are assigned to a flight whose number ends with '1'.

```
sql SELECT c.CABINCREW_ID, c.FIRST_NAME, c.LAST_NAME, c.CONTACT, f.FLIGHT_ID FROM cabincrew c JOIN flight f ON c.FLIGHT_ID = f.FLIGHT_ID WHERE c.FIRST_NAME LIKE 'A%' AND f.FLIGHT_ID LIKE '%1';
```

Q3.20 Total passengers and total baggage items for flights arriving in Paris on 11 Feb 2024.

```
sql SELECT f.FLIGHT_ID, COUNT(bp.PASSENGER_ID) AS Total_Passengers, SUM(bp.BAGGAGE) AS Total_Baggage FROM flight f JOIN boardingpass bp ON f.FLIGHT_ID = bp.FLIGHT_ID WHERE f.FLIGHT_TO = 'Paris' AND f.ARRIVAL_DATE = '2024-02-11' GROUP BY f.FLIGHT_ID;
```

Q3.21 Number of women's products available, alias `product_count`.

```
sql SELECT COUNT(*) AS product_count FROM product p JOIN product_category pc ON p.PRODUCT_ID = pc.PRODUCT_ID JOIN category c ON pc.CATEGORY_ID = c.CATEGORY_ID WHERE c.NAME = 'Women';
```

Q3.22 Train name and type for trains with speed less than 50.

```
sql SELECT train_name AS TRAIN_NAME, train_type AS TRAIN_TYPE FROM train_details_tbl WHERE train_speed < 50;
```

Q3.23 Distinct first names and contact numbers of passengers departing from 'Hong Kong', on flight ID 4, who requested a 'Vegetarian' meal.

```
sql SELECT DISTINCT p.FIRST_NAME, p.CONTACT FROM passenger p JOIN boardingpass bp ON p.PASSENGER_ID = bp.PASSENGER_ID JOIN flight f ON bp.FLIGHT_ID = f.FLIGHT_ID WHERE f.FLIGHT_FROM = 'Hong Kong' AND bp.FLIGHT_ID = 4 AND bp.MEAL = 'Vegetarian';
```

Q3.24 Product id and name of products whose status is 'In the transit hub'.

```
sql SELECT DISTINCT p.PRODUCT_ID, p.NAME FROM product p JOIN order_item oi ON p.PRODUCT_ID = oi.PRODUCT_ID JOIN order_delivery od ON oi.ORDER_DELIVERY_ID = od.ORDER_DELIVERY_ID WHERE od.STATUS = 'In the transit hub';
```

Q3.25 Artist id and name where the artist has a **number** in the name.

```
sql SELECT ARTIST_ID, NAME FROM artist WHERE NAME REGEXP '[0-9]';  
Fallback if REGEXP is unsupported: ten OR'd NAME LIKE '%0%' ... '%9%' conditions.
```

Q3.26 Message id and content where the message contains "Hello".

```
sql SELECT MESSAGE_ID, CONTENT FROM message WHERE CONTENT LIKE '%Hello%';
```

Q3.27 Full name, license number and plate number of drivers whose vehicles are "In Use" and whose plate number ends with "0".

```
sql SELECT CONCAT(d.FIRST_NAME, ' ', d.LAST_NAME) AS Name, d.LICENSE_NUMBER, v.PLATE_NUMBER FROM driver d JOIN vehicle v ON d.DRIVER_ID = v.DRIVER_ID WHERE v.STATUS = 'In Use' AND v.PLATE_NUMBER LIKE '%0';
```

Q3.28 License number, vehicle id, rating and booking id of drivers with rating ≥ 4.5 whose booking status is other than "Cancelled".

```
sql SELECT d.LICENSE_NUMBER, v.VEHICLE_ID, d.RATING, b.BOOKING_ID FROM driver d JOIN vehicle v ON d.DRIVER_ID = v.DRIVER_ID JOIN booking b ON v.VEHICLE_ID = b.VEHICLE_ID WHERE d.RATING >= 4.5 AND b.STATUS <> 'Cancelled';  
<> (or !=) for "other than". Do not write NOT b.STATUS = ...
```

Q3.29 Number of viewers having the same name.

```
sql SELECT viewername, COUNT(*) AS name_count FROM viewer GROUP BY viewername;
```

Q3.30 Full name (alias FULLNAME) of users who have a contact with any kind of engineer (any job title containing "Engineer").

```
sql SELECT DISTINCT CONCAT(u.FIRST_NAME, ' ', u.LAST_NAME) AS FULLNAME FROM users u JOIN contacts c ON u.USER_ID = c.USER_ID JOIN users cu ON c.CONTACT_ID = cu.USER_ID JOIN jobs j ON cu.USER_ID = j.USER_ID WHERE j.JOB_TITLE LIKE '%Engineer%';
```

The hardest query in the set — a self-join: `users` appears twice (once as the person, once as their contact). Be able to reproduce this from memory.

Stages 4-6 — Coding Round (68 problems)

Source: *Accenture Coding.pdf*. Re-ordered from alphabetical into difficulty tiers.

Read this before you start. The original PDF lists these alphabetically, which is the worst possible practice order — it puts "Autobiographical Number" (hard) third and "Reverse String" (trivial) forty-seventh. Below they are tiered.

Under each problem is the **pattern** it teaches, not the full code. Write the code yourself; that is the entire point. Target times: Tier 1 ≤ 10 min, Tier 2 ≤ 20 min, Tier 3 ≤ 30 min.

Universal rules for the Accenture coding round

- Read the **Notes** block. Almost every problem has a "return -1 if null / return 0 if length < 3" clause and the hidden test cases check exactly that.
- Print **exactly** what is asked. No "Enter a number:" prompts in the final submission.
- Array indices start at 0 unless stated; "even position" usually means index 0, 2, 4.
- Handle empty input, single-element input, and all-same-element input every single time.

Stage 4 — Tier 1: Warm-up (24 problems)

Loops, conditionals, digits, one-pass array/string scans. If any of these takes you more than 10 minutes, stop and drill it until it doesn't.

C1. Binary to Decimal — Input `1010` → `10`.

Pattern: positional weights, or `result = result*2 + digit`.

C2. Decimal to Binary — Input `10` → `1010`.

Pattern: repeated `n % 2 / n /= 2`, collect remainders, reverse.

C3. Table — Print the table of a number and the sum of all its multiples. Input `5` → `5,10,...,50` and `275`.

Pattern: single loop 1..10, accumulate.

C4. Palindrome Numbers in a Range — Lower and upper limit, print all palindromes between. `10, 80` → `11,22,...,77`.

Pattern: reverse-the-number helper inside a range loop.

C5. Googly Prime Number — A number whose **digit sum** is prime. `43` → YES (4+3=7); `123` → NO (sum 6).

Pattern: digit-sum helper + isPrime helper. Two small functions, composed.

C6. Sum of Primes less than N — `10` → `17` (2+3+5+7).

Pattern: isPrime in a loop; trial division to $\sqrt{n}$ is enough at this size.

C7. Sum of Divisors — `12` → `28` (1+2+3+4+6+12).

Pattern: loop 1..n and test `n % i == 0`. Optimisation: loop to $\sqrt{n}$ and add both divisors.

C8. Sum of Numbers Divisible by both 3 and 5 — Between m and n inclusive. `12, 50` → `90` (15+30+45).

Pattern: divisible by both 3 and 5 means divisible by **15**. Say that out loud before you write the loop.

C9. Fibonacci Series — Print the series up to n terms.

Pattern: two rolling variables. No recursion — it will time out on large n.

C10. Nth Fibonacci — `9` → `34` (0-indexed series `0,1,1,2,3,5,8,13,21,34`).

Pattern: same loop. **Confirm whether the series is 0-indexed or 1-indexed before submitting** — that is the actual trap.

C11. Roots of a Quadratic Equation — `ax2 + bx + c = 0`.

Pattern: discriminant `d = b2-4ac`. Three branches: $d > 0$ (two real), $d = 0$ (one), $d < 0$ (imaginary — print the complex form, don't crash).

C12. Standard Deviation — of a list of numbers.

Pattern: two passes — mean, then mean of squared deviations, then square root.

C13. Sum of Distances Between Three Points — For (1,1), (2,4), (3,6), distance = $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$, summed over all three pairs.

Pattern: a distance helper called three times. Watch the source PDF's formula, which drops a bracket — the correct form is $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$.

C14. Operation Choices — `int OperationChoices(int c, int n, int a, int b)`: return a+b if c=1, a-b if c=2, a*b if c=3, a/b if c=4.

Pattern: switch statement. Free marks — but guard division by zero.

C15. Maximum in Array — `void MaxInArray(int arr[], int length)`: print the maximum element and its index on **separate lines**. 23 45 82 27 66 12 78 13 71 86 → 86 then 9.

Pattern: single pass tracking max and its index. Note the strict output format.

C16. Maximum With Index — [1,8,4,9,6] → (9,3). Same as above, different return shape.

C17. Second Largest Element in an array.

Pattern: one pass with two variables (largest, second). Do **not** sort — and handle duplicates of the largest.

C18. Missing Number — [1,2,4,5,6] → 3.

Pattern: expected sum $n(n+1)/2$ minus actual sum. O(n), no sorting.

C19. Remove Duplicates — [1,2,2,3,4,4,5] → [1,2,3,4,5].

Pattern: HashSet, or two-pointer if the array is sorted.

C20. Count Occurrences of Each Element — {10,5,10,15,10,5} → 10-3, 5-2, 15-1.

Pattern: HashMap frequency count. **This is the single most reusable pattern in the whole set** — anagram, most frequent vowel, and rhyme words all reduce to it.

C21. Reverse a String.

Pattern: two pointers swapping inward, or build backwards. Know how to do it without the library `reverse()`.

C22. Repeat a String — N = 3, S = "abc" → "abcabcabc".

Pattern: StringBuilder in a loop. Never += a String inside a loop in Java.

C23. Length of Last Word — " I am a passionate Developer " → 9.

Pattern: trim trailing spaces first, then scan backwards to the first space. The trailing whitespace **is** the test case.

C24. First K Words — "Hello I am a passionate developer", k=4 → "Hello I am a".

Pattern: split on whitespace, take the first k, re-join with single spaces.

C25. Print Even Odd — [1,2,3,4,5,6] → "odd even odd even odd even".

Pattern: build a result string from $n \% 2$. Mind the single space separator and no trailing space.

Stage 5 — Tier 2: Core Patterns (26 problems)

Hashing, two pointers, sorting, bit manipulation, in-place array work. This tier is where the round is actually won.

C26. Anagram — Can the characters of s be rearranged to form t? "listen"/"silent" → True.

Pattern: compare sorted strings, or compare 26-slot frequency arrays (O(n), preferred). Check lengths first and return early.

C27. Most Frequent Vowel — Find the vowel occurring most often in a string. xyuaab → a.

Pattern: frequency count restricted to `aeiou` . [HIGH-FREQ — appears twice in the source PDF as #34 and #63]

C28. Longest Word in a list of strings — "yes no number" → "number".

Pattern: single pass tracking max length. Decide the tie-breaking rule (first occurrence) before you code.

C29. Reverse Words — "Hello World" → "World Hello".

Pattern: split, reverse the array, join. Not the same as reversing characters — read carefully.

C30. Rhyme Words — Group or find words that share an ending.

Pattern: compare the last k characters; group with a HashMap keyed on the suffix.

C31. Replace Character — Swap all `ch1` with `ch2` and all `ch2` with `ch1` . "apples", a↔p → "paales".

Pattern: a single pass with an if/else-if chain. The classic bug is doing two sequential replaces, which converts everything into one character.

C32. Move Hyphens to Front — "Move-Hyphens-to-Front" → "---MoveHyphenstoFront".

Pattern: count hyphens, then emit that many followed by the non-hyphen characters. A stable-partition problem.

C33. Password Checker — Return 1 if valid: ≥ 4 characters, ≥ 1 digit, ≥ 1 capital, no space or `/` , and the first character is not a digit.

Pattern: five boolean flags set in one pass, ANDed at the end. Build a checklist — partial credit here is common.

C34. Replace Most Frequent Characters — Find the most frequent character(s) and replace them.

Pattern: frequency map → find max → second pass to rebuild the string.

C35. Absolute Difference — Count elements of `arr` whose absolute difference with `num` is $\leq$ `diff` . `[12,3,14,56,77,13]` , `num=13`, `diff=2` → `3` . **Return -1 if none.**

Pattern: one pass with `Math.abs` . The `-1` case is the hidden test.

C36. Count Carry — Number of carries generated when adding two numbers digit by digit, right to left. `451 + 349` → `2` .

Pattern: simulate column addition; `carry = (d1+d2+carry) / 10` .

C37. Encode Number — Square each digit and concatenate. `34` → `916` (9 and 16).

Pattern: extract digits, square, append to a string, parse back to int at the end.

C38. Sum of Binary Digits — Convert n to binary and sum the digits. `15` (`1111`) → `4` .

Pattern: this is a **popcount** — `while(n>0){count += n&1; n >>= 1;}` .

C39. Magical Numbers from 1 to N — Convert to binary, map 0→1 and 1→2, sum the digits; magical if the sum is odd. `N=5` → `2` .

Pattern: for a b-bit number with s set bits, the sum is $(b - s)*1 + s*2 = b + s$. Derive that and the problem collapses to a one-liner.

C40. Rearrangement of Bits — Rearrange the set bits of N to be consecutive and as small as possible. `10` (`1010`) → `3` (`0011`).

Pattern: count set bits k, answer = $(1 \ll k) - 1$.

C41. Max Exponent of 2 — Return the number in `[a,b]` with the highest power-of-2 factor; on a tie, return the smaller. `7..12` → `8` .

Pattern: for each i, count how many times 2 divides it. Strict `>` in the comparison preserves the smaller number on ties.

C42. Binary Operations String — `1C0C1C1A0B1` where A=AND, B=OR, C=XOR, evaluated strictly left to right with no precedence. → `1` . **Return -1 if the string is null.**

Pattern: scan in steps of 2, apply the operator to the running result. Explicitly *ignoring* precedence is the trick.

C43. Equilibrium Sum — Index where the sum to the left equals the sum to the right. `{3,4,3,1,6}` → `2`. Return `-1` if none.

Pattern: total sum, then one pass maintaining `leftSum`; check `leftSum == total - leftSum - arr[i]`. $O(n)$, not $O(n^2)$.

C44. Sum at Even Indices of the Reversed Array — `[10,20,30,40,50,60]` → `120`.

Pattern: you never need to reverse. Index `i` of the reversed array is `n-1-i` of the original. Reversing first is fine too, but see the trick.

C45. Negative Stock Price Days — Count days where the closing price fell. `{2,3,1,4,5,2}` → `2`.

Pattern: compare each element with its predecessor. Single pass.

C46. Maximum Difference Between Successive Elements (sorted form) — `[3,6,9,1]` → `3`. Return `0` if fewer than 2 elements.

Pattern: sort, then one pass of adjacent differences.

C47. Merge Two Sorted Arrays — `[1,2,3,4,5]` and `[2,4,6,8,10]` → `[1,2,2,3,4,4,5,6,8,10]`.

Pattern: two pointers, the merge half of merge sort. Do **not** concatenate and sort — they are testing the merge.

C48. Intersection of Two Arrays — Elements present in both.

Pattern: HashSet of the first array, then filter the second. Decide whether duplicates should appear once or repeatedly.

C49. Rotate Array by K — `[1,2,3,4,5,6,7]`, `k=3` → `[5,6,7,1,2,3,4]`.

Pattern: the **reversal algorithm** — reverse the whole array, reverse the first `k`, reverse the rest. $O(n)$ time, $O(1)$ space. Remember `k = k % n`.

C50. Elevation Point — `[1,2,3,4,3,2,1]` → `4`; `[5,3]` → `5`.

Pattern: find the peak — the element greater than both neighbours. Handle the case where the peak is at an end.

C51. Large-Small Sum — Return (second largest among elements at **even** positions) + (second smallest among elements at **odd** positions). `3 2 1 7 5 4` → `7`. Return `0` if the array is empty or length ≤ 3 .

Pattern: split by index parity into two lists, sort each, pick. **[HIGH-FREQ — this appears three times in the source PDF, as #21, #27 and #53, each with a slightly different wording of which element to take from which list.]** Read the exact wording each time.

C52. Product of Smallest Pair — Find the two least elements whose sum $\leq$ `sum`, return their product. `sum=9`, `5 2 4 3 9 7 1` → `2`. Return `-1` if empty or `n < 2`; `0` if no such pair.

Pattern: sort, take the first two, verify the constraint. Watch negatives: `9 8 3 -7 3 9` with `sum=4` → `-21`.

C53. Rat Count House — Rats `r` each eat `unit` food; `arr[i]` is the food in house `i+1`. Return how many houses (from the first) are needed to feed all rats. `r=7`, `unit=2`, `2 8 3 5 7 4 1 2` → `4`.

Pattern: running prefix sum until it reaches `r*unit`. Return `-1` if the array is null, `0` if the total is insufficient.

Stage 6 — Tier 3: Hard (18 problems)

Sliding windows, in-place matrix work, linked lists, and problems where the naive solution is too slow or simply wrong. Do these last, but do them.

C54. Longest Substring Without Repeating Characters — "abcabcbb" → `3`.

Pattern: **sliding window** with a HashMap of last-seen indices. The $O(n^2)$ brute force will pass small tests and fail the large ones. This is the most classic interview problem in the set — learn it properly.

C55. Max Favourite Song — Given string `S` and window size `K`, find the maximum count of 'a' in any substring of length `K`. "acdbaaca", `K=3` → `2`.

Pattern: **fixed-size sliding window** — add the entering character, remove the leaving one. Do not rebuild the count each time.

C56. Chocolate Distribution — Choose `m` packets from `n` so that $(\max - \min)$ among the chosen is minimum. `{7,3,2,4,9,12,56}`, `m=3` → `2`.

Pattern: sort, then slide a window of size m and track $arr[i+m-1] - arr[i]$. The insight is that the chosen packets must be consecutive after sorting.

C57. Pair Sum with Maximum Product — Among all pairs summing to target, return the one with the largest product (larger value first). target=18, $[11, 1, 2, 8, 10, 11, 15, 7] \rightarrow [10, 8]$.

Pattern: sort + two pointers converging from both ends. Note the maths: for a fixed sum, the product is maximised when the two values are closest together.

C58. Set Matrix Zeroes — If an element is 0, zero its whole row and column. $[[1, 1, 1], [1, 0, 1], [1, 1, 1]] \rightarrow [[1, 0, 1], [0, 0, 0], [1, 0, 1]]$.

Pattern: two marker arrays (one for rows, one for columns), then a second pass. **The trap is mutating in place during the first pass, which cascades zeroes across the entire matrix.** The $O(1)$ -space version uses the first row and column as the markers.

C59. Linked List Palindrome — $[1, 2, 2, 1] \rightarrow \text{true}$.

Pattern: slow/fast pointers to find the middle, reverse the second half, compare the halves. You must be able to write `reverse(Node head)` from memory.

C60. Autobiographical Number — The digit at position i counts how many i 's appear in the number. "1210" $\rightarrow$ return 3 (the count of distinct digits). Return 0 if not autobiographical or if the string is None.

Pattern: frequency count, then verify $freq[i] == digit[i]$ for every position. Then count distinct digits. Two separate sub-problems glued together — that gluing is the difficulty.

C61. Bulb Switch — Pressing switch i toggles bulb i and **every bulb to its right**. Find the minimum presses to turn all bulbs on. $[0, 1, 0, 1] \rightarrow 4$; $[1, 0, 0, 0, 0] \rightarrow 1$.

Pattern: scan left to right tracking how many flips have occurred so far; the effective state of bulb i is $arr[i] \text{ XOR } (flips \% 2)$. Press whenever the effective state is 0. Greedy, $O(n)$.

C62. String Decoder — Decode a sequence of 1s and 0s into a word, where each uppercase letter is represented by a run of 1s. $10110111 \rightarrow \text{ABC}$.

Pattern: split on the 0 delimiters, count the 1s in each group, map count $\rightarrow 'A' + \text{count} - 1$.

C63. Vowel Permutation — Count permutations of a string formed by keeping the vowels fixed in place. "ABC" $\rightarrow 2$.

Pattern: count the consonants c ; the answer is $c!$. Factorials overflow fast — note the expected return type.

C64. Matrix Even/Odd Second Largest — Split the array into even-index and odd-index sub-arrays, sort both ascending, print the sum of the second largest from each.

Pattern: same skeleton as C51 — see the note there about the three near-identical variants. Guard against a sub-array of size < 2 .

C65. Small-Large Sum — Second largest from even positions + second largest from odd positions. $3\ 2\ 1\ 7\ 5\ 4 \rightarrow 7$. Return 0 if the array is empty or length ≤ 3 .

Pattern: the third variant of the family. Diff it against C51 and C64 and note which asks for *smallest*.

C66. Find Count (Absolute Difference) with constraints — see C35; the harder variant adds a sorted-input requirement and asks for an $O(\log n)$ solution.

Pattern: binary search for the boundaries $num - diff$ and $num + diff$; the answer is the index gap.

C67. Maximum Element and Index in a single traversal, with ties — Extend C15/C16 to report all indices where the maximum occurs.

Pattern: collect into a list; reset the list when a new maximum is found.

C68. Vowel Repetition — Identical to C27 in the source PDF (#63 duplicates #34). Use it as a timed re-test rather than as new material.

Coding-round checklist

Before you submit anything:

- ☐ Null / empty input handled with the exact return value the Notes demand
 - ☐ Single-element and two-element arrays
 - ☐ All elements identical
 - ☐ Negative numbers (C52 in particular)
 - ☐ Integer overflow on factorials and concatenations
 - ☐ Output format is exactly as specified — no prompts, no trailing spaces, correct newlines
 - ☐ `k % n` before any rotation
 - ☐ 0-indexed vs 1-indexed confirmed against the worked example
-

Stage 7 — OOP & Programming Concepts

Source: *Computer Science.pdf*. Short section, but it repeats almost verbatim year to year.

Q7.1 Which concept is used here?

```
class A {  
    void sum() { ... }  
    void sum(int a) { ... }  
}
```

(a) Encapsulation (b) Inheritance (c) Polymorphism (d) Abstraction

Answer: (c) Polymorphism. Two methods with the same name but different parameter lists inside one class is **method overloading** — compile-time (static) polymorphism.

Q7.2 Which type of polymorphism does this show?

```
public class TeaMac {  
    public Coffee boilTea(ChooseTea choosed) { ... }  
    public List boilTea(ChooseTea choosed) { ... }  
}
```

(a) Static polymorphism (b) Dynamic polymorphism (c) Late binding (d) Method overriding

Answer: (a) Static polymorphism. Overloading is resolved by the compiler. Overriding (a subclass redefining a parent method) is dynamic/runtime polymorphism, resolved by the JVM. **[HIGH-FREQ]** — know the static vs dynamic split cold, it is asked in two different forms.

Q7.3 What is the output?

```
class Test {  
    public int x = 45;  
    public int y;  
    public double z;  
    public void F1(int a) { y = a; }  
    public double F2() { return 5; }  
}  
// main: t.x = 9; t.y = 18; t.z = 3; t.F1(21);  
System.out.println(t.F2());
```

(a) 3 (b) 5 (c) 9 (d) 18

Answer: (b) 5. `F2()` returns the literal 5 and never reads a field. Every assignment before it is a distractor. **[HIGH-FREQ — asked twice, as Q7 and Q14 of the source paper.]**

Q7.4 What is the output?

```
class Children {  
    private final void flipper() { System.out.println("Children"); }  
}  
public class Parent extends Children {  
    public final void flipper() { System.out.println("Parent"); }  
    public static void main(String[] args) { new Parent().flipper(); }  
}
```

(a) Children (b) Parent (c) Children then Parent (d) Error

Answer: (b) Parent. A `private` method is **not inherited**, so `Parent.flipper()` is not an override — it is an unrelated method. The call resolves to Parent's version.

Q7.5 Which statement is true?

```
class Employee extends Person {  
    private Address address;  
}
```

Answer: Employee has an **is-a** relationship with Person (inheritance, via `extends`) and a **has-a** relationship with Address (composition, via a field). "extends → is-a, field → has-a" is the whole rule.

Q7.6 What is the output?

```
var my_arr = [3, 6, 2, 18];  
function myFunction(num) { return (num + 5); }  
document.write(my_arr.map(myFunction));
```

(a) 3,6,2,18 (b) 8,11,7,23 (c) false (d) error

Answer: (b) 8,11,7,23. `map()` applies the function to every element and returns a new array of the same length. Contrast with `filter()` (keeps a subset) and `reduce()` (collapses to one value).

Q7.7 For an application that must run across many different hardware configurations and operating systems, which OS property matters most?

Answer: the two source papers disagree — one says **Resource allocation**, the other says **Device management**. Both are defensible: device management is the OS's ability to talk to varied hardware; resource allocation is fair distribution of CPU/memory. **Pick "Device management" when the question stresses *varying hardware*, "Resource allocation" when it stresses *smooth execution across systems*.** See the errata.

Stage 8 — Operating Systems & Computer Architecture

Q8.1 A program must create a new process and run a different executable in a Unix-like OS. Which pair of system calls?

- (a) create() and run() (b) fork() and exec() (c) fork() and wait() (d) spawn() and terminate()

Answer: (b) fork() and exec(). `fork()` duplicates the current process; `exec()` replaces the child's program image with the new executable. `wait()` only makes the parent block. **[HIGH-FREQ]**

Q8.2 During a context switch, which CPU register holding the address of the next instruction must be saved and restored?

- (a) Stack Pointer (b) General Purpose Register (c) Program Counter (d) Instruction Register

Answer: (c) Program Counter. The PC holds the address of the next instruction to fetch; without saving it, the process cannot resume where it stopped. The Instruction Register holds the instruction *currently* being decoded — a deliberate distractor. **[HIGH-FREQ — appears three times across the papers.]**

Q8.3 You are designing a CPU control unit and must specify how data moves from the accumulator to memory. What is this operation called?

- (a) Register transfer (b) Memory read (c) Memory write (Store) (d) Instruction fetch

Answer: (c) Memory write / Store. CPU → memory is a write; memory → CPU is a read. **[HIGH-FREQ — three occurrences.]**

Q8.4 I/O devices are attached by wires; the host controller sends messages to a device controller that performs the operation. Which device do you choose if the controllers have a **built-in cache** for faster transfer?

- (a) Enhanced Interface Drive Electronics (b) **Enhanced Integrated Drive Electronics (EIDE)** (c) Enhanced Interface Driver Electronics (d) Enhanced Integrated Hard Driver

Answer: (b) EIDE. An enhanced version of IDE/ATA for connecting storage devices, with controller functionality and caching built in. The three distractors are near-anagrams — read every word. **[HIGH-FREQ]**

Q8.5 Data is transferred between CPU and a peripheral over a shared bus; the system uses controllers with **dedicated high-speed buffers**. Which controller is most suitable?

- (a) Wireless (b) USB (c) SATA (d) Ethernet

Answer: (c) SATA. Serial ATA is purpose-built for high-speed storage transfer. Note how close this is to Q8.4 — the distinguishing words are "built-in cache" (EIDE) versus "high-speed buffers on a shared bus" (SATA).

Q8.6 Which physical component on the motherboard is the primary pathway for data, addresses and control signals between CPU, memory and I/O?

- (a) PCI slot (b) System Bus (c) CPU Register (d) Northbridge/Southbridge chipset

Answer: (b) System Bus — comprising the Data Bus, Address Bus and Control Bus.

Q8.7 After a security breach, which OS property should an IT manager prioritise?

- (a) Access control (b) User interface (c) Multitasking (d) Virtual memory

Answer: (a) Access control. It decides who may reach which resource and perform which operation. **[HIGH-FREQ]**

Q8.8 Besides creating, deleting and manipulating files and directories, which additional function most improves a file-management subsystem's ability to organise and access files efficiently?

- (a) File compression (b) Managing file permissions and access control (c) File encryption (d) Backup and recovery

Answer: (b) Permissions and access control. Compression, encryption and backup are storage/security add-ons; access control is core to structured multi-user file management.

Q8.9 A cloud infrastructure must run multiple isolated environments on one physical server. Which technology?

- (a) Containers (b) Virtual Machines (c) Serverless (d) Bare metal

Answer: (b) Virtual Machines. A hypervisor gives each VM its own OS and strong isolation. **[HIGH-FREQ]** — see Q12.4 for the mirror-image question where containers are the answer.

Q8.10 Which technology allows multiple operating systems to run on a single physical machine?

(a) Virtualization (b) Containerization (c) Clustering (d) Replication

Answer: (a) Virtualization. The hypervisor abstracts the hardware. Containers share the host kernel and therefore cannot run a *different* OS.

Q8.11 A containerized application is consuming excessive memory, causing performance problems. Recommended first approach?

(a) Increase the memory limit (b) Use a larger base image (c) Reduce replicas (d) Analyze the application code for memory leaks

Answer: (d) Analyze for memory leaks. Raising the limit hides the symptom. **[HIGH-FREQ]** — note the pattern: when a question offers "add more resources" versus "find the root cause", the exam always wants root cause.

Q8.12 A containerized application shows high latency. What should the engineer do **first**?

(a) Increase CPU and memory limits (b) Analyze network traffic and logs (c) Restart the container (d) Reduce the number of containers

Answer: (b) Analyze traffic and logs. Same principle as Q8.11 — diagnose before you change anything.

Q8.13 An application relies on multiple hardware peripherals and needs strict isolation from other containers. What is the most critical factor in maintaining operational security and preventing resource contention?

Answer: Configuring the **kernel** to enforce strict resource isolation and limit peripheral access to only the allocated devices. **[HIGH-FREQ — this exact question appears three times across the papers with slightly reworded options.]** The kernel controls hardware access, so isolation must be enforced there, not in the scheduler or the virtualization layer.

Stage 9 — Web Technologies & APIs

Q9.1 A mobile app retrieves a list of restaurants from a food-delivery API and must only read data. Which HTTP method?
(a) POST (b) PUT (c) DELETE (d) GET

Answer: (d) GET. Safe and idempotent — retrieves without modifying server state. **[HIGH-FREQ]**

Q9.2 An app sends data to a server using an API via POST. What is the primary characteristic of POST here?

Answer: It submits data to be processed by the server, typically creating a new resource. GET retrieves, PUT updates/replaces, PATCH partially updates, DELETE removes.

Q9.3 A developer hits an incorrect or unavailable API endpoint and gets "Not Found". Which status code?
(a) 404 (b) 500 (c) 403 (d) 400

Answer: (a) 404. Learn the full family: **200** OK, **201** Created, **301** Moved Permanently, **400** Bad Request, **401** Unauthorized, **403** Forbidden, **404** Not Found, **500** Internal Server Error, **502** Bad Gateway, **503** Service Unavailable. **[HIGH-FREQ]**

Q9.4 A REST endpoint must retrieve the details of a single product by its unique ID/SKU. Which method and URL structure?
(a) `POST /products/{id}` (b) `GET /products/{id}` (c) `PUT /products/{id}` (d) `DELETE /products/{id}`

Answer: (b) `GET /products/{id}`. REST names resources with **nouns**, identifies a specific one with a path parameter, and expresses the action through the HTTP verb. Never `/getProduct` or `/products/{id}/details`. **[HIGH-FREQ]**

Q9.5 A public API has breaking changes to endpoints and data structures. Which versioning strategy is generally recommended?
(a) Modify existing endpoints directly (b) Query parameter `?version=2` (c) Version in the URL path, e.g. `/api/v2/resource` (d) Custom HTTP headers

Answer: (c) Version in the URL path. Explicit, cacheable, visible in logs. Header versioning is valid in practice but is not the answer the exam wants.

Q9.6 Configuring Amazon S3 bucket-level CORS: which key generates a CORS preflight request to HTML?

Answer: `corsconfiguration=enabled`. Understand the concept behind it: a **preflight** is the browser's automatic **OPTIONS** request, sent before a cross-origin request that isn't "simple", asking the server which origins, methods and headers it permits.

Q9.7 What is the output when the user clicks the button?

```
<button onclick="myFunction(this);">Test onclick attribute</button>
<script>
function myFunction(button) {
    document.write(button.getAttribute("onclick", 1));
}
</script>
```

Answer: `myFunction(this);` — `getAttribute` returns the **literal string assigned to the attribute in the HTML, not the executed result.**

Q9.8 What is the output?

```
<button onclick="myFunction(this);">Test attribute</button>
<script>
function myFunction(button) {
    var newAttr = document.createAttribute("myAttribute");
    newAttr.value = "My Value";
    button.setAttributeNode(newAttr);
    document.write(button.getAttribute("myAttribute"));
}
</script>
```

Answer: `My Value` . `createAttribute` → `set .value` → `setAttributeNode` attaches it → `getAttribute` reads it back.

Q9.9 What does the text inside this canvas tag indicate?

```
<canvas id="myCanvas">HTML5 canvas not supported</canvas>
```

(a) A comment for the user (b) A comment for incompatible browsers (c) A comment for IE and Opera (d) All of the above

Answer: (b). It is **fallback content**, rendered only by browsers that do not support `<canvas>` . The same pattern applies to `<video>` and `<audio>` .

Q9.10 When is `myFunction()` called?

```

```

(a) When the page is closed (b) When the user clicks Esc before the page loads (c) When the page is refreshed (d) On load

Answer: (b). `onabort` fires when the loading of an object is aborted before completing — pressing Esc mid-load does exactly that.

Q9.11 Which code makes a block turn circular on hover and revert on mouse-out, with a slow-fast-slow animation?

Answer: `border-radius: 50px` together with `transition: 0.5s` . `border-radius` rounds the corners (50% or a radius $\geq$ half the side makes it a circle); `transition` supplies the smooth easing. Default easing is `ease` , which is exactly slow-fast-slow.

Q9.12 Which code is appropriate for autoplaying audio?

Answer:

```
html <audio autoplay> <source src="test.mp3"> <source src="test.ogg"> </audio>
```

The `autoplay` attribute goes on `<audio>` ; multiple `<source>` children let the browser choose a format it supports.

Q9.13 Which protocol secures data **in transit** over the internet by encrypting communication between systems?

(a) HTTP (b) FTP (c) SSL/TLS (d) SMTP

Answer: (c) SSL/TLS. TLS is the modern standard (SSL is deprecated but still the exam's wording). It is what makes HTTP into HTTPS. **[HIGH-FREQ]**

Q9.14 A company wants secure and reliable **email transmission** for its cloud-based email service. Which protocol?

(a) HTTP (b) FTP (c) SMTP (d) POP3

Answer: (c) SMTP — used to **send** mail. IMAP and POP3 **retrieve** it (IMAP syncs and keeps mail on the server; POP3 downloads and typically deletes). **[HIGH-FREQ]**

Stage 10 — DBMS Concepts

Q10.1 Transaction A does `Stock = Stock - 1`, Transaction B does `Stock = Stock - 2`, both on ProductID 1001. Both succeed, but the final stock reflects only one update. Which concurrency issue?

(a) Phantom Read (b) Lost Update (c) Non-repeatable Read (d) Deadlock

Answer: (b) Lost Update. Both read the same original value; the second write overwrites the first. Know all four: **Dirty read** (reading uncommitted data), **Non-repeatable read** (same row, different value on re-read), **Phantom read** (same query, new rows appear), **Lost update** (as here).

Q10.2 Employees are uniquely identified by EmployeeID but may change PhoneNumber. Which column should be the Primary Key?

(a) Name (b) PhoneNumber (c) EmployeeID (d) Department

Answer: (c) EmployeeID. A primary key must be **unique, non-null and immutable**. PhoneNumber fails immutability; Name and Department fail uniqueness.

Q10.3 See Q2.7 — indexing a high-cardinality column queried by equality is the standard performance fix.

Q10.4 See Q2.8 — `AFTER DELETE` triggers and the `OLD` / `NEW` row references.

Stage 11 — Networking & Security

Sources: *Fundamentals of Networking.pdf* + *Actual Paper MCQs.pdf*. The largest theory section, and the most repetitive — learn the taxonomies, not the individual questions.

OSI model

Q11.1 A user connects a computer, smartphone and smart TV to a router. Which OSI layer transmits the Ethernet, cable and Wi-Fi signals?

Answer: Physical Layer (L1) — raw bit streams over copper, fibre and radio. **[HIGH-FREQ]**

Q11.2 Data packets move between routers and switches across network segments. Which layer establishes logical paths and routes packets?

Answer: Network Layer (L3) — logical (IP) addressing and routing. Routers live here. **[HIGH-FREQ]**

Q11.3 During a video call, which layer is responsible for error checking, ordering and reliable delivery end to end?

Answer: Transport Layer (L4) — TCP's segmentation, sequencing, acknowledgement and flow control.

Memorise the full stack, bottom to top: Physical → Data Link (MAC, switches, frames) → Network (IP, routers, packets) → Transport (TCP/UDP, ports, segments) → Session → Presentation (encryption, compression) → Application (HTTP, SMTP, DNS).
Mnemonic: *Please Do Not Throw Sausage Pizza Away*.

Protocols and topology

Q11.4 A service needs fast, connectionless query-response where packet loss is acceptable but retransmission delay is not. Which protocol?

(a) HTTP (b) FTP (c) DNS (d) SMTP

Answer: (c) DNS, which runs primarily over **UDP**. HTTP, FTP and SMTP all sit on TCP. The underlying point is TCP (reliable, ordered, handshake) versus UDP (fast, connectionless, lossy).

Q11.5 A client sends a SYN packet to a server. What does an available server respond with?

Answer: SYN-ACK. The full TCP three-way handshake is **SYN → SYN-ACK → ACK**.

Q11.6 A small business uses a ring topology. If a single workstation goes down, what happens?

(a) Minimal impact, traffic reroutes (b) Network becomes isolated (c) Network slows down (d) Network breaks completely

Answer: (d) Network breaks completely (in a traditional single ring — a dual ring can survive it). Compare: **star** (one node fails, only that node is affected; the switch is the single point of failure), **bus** (a break in the backbone kills the segment), **mesh** (highly fault-tolerant). **[HIGH-FREQ]**

Q11.7 Which method prevents unauthorised devices from connecting to a network?

Answer: MAC filtering — an allow-list of hardware addresses. Weak on its own (MACs are spoofable) but it is the exam's answer.

Q11.8 Which is the most secure wireless encryption protocol for Wi-Fi?

Answer: WPA3. Order of strength: WEP (broken) < WPA < WPA2 < WPA3. **[HIGH-FREQ]**

Q11.9 A developer configures WPA3-Enterprise on Android with `CONFIG_SAE=y`, `CONFIG_SUITEB=y`, `CONFIG_SUITEB192=y`. Which line is unnecessary or weakening?

Answer: `CONFIG_SAE=y` is unnecessary — SAE (Simultaneous Authentication of Equals) belongs to WPA3-**Personal**. WPA3-Enterprise authenticates with 802.1X/EAP. (One source paper names `SUITEB192` instead; see the errata.)

Attacks

Q11.10 Hackers steal data by **specifying the route the packets should take**, bypassing a packet-filter firewall that does not analyse source information. What attack is this?

Answer: Source Routing Attack. The sender dictates the path instead of letting routers choose it. **[HIGH-FREQ]**

Q11.11 Attackers intercept packets and **inject malicious routing information** so traffic is redirected through their own systems, letting them eavesdrop or alter data.

Answer: Routing Table Poisoning. (One paper labels an almost identical question "Man-in-the-Middle" — MITM is the umbrella category, routing table poisoning is the specific technique. If both appear as options, choose the specific one.) **[HIGH-FREQ]**

Q11.12 A core web server is hit by an overwhelming flood of bot traffic mixed with legitimate traffic. Most effective immediate solution?

Answer: DDoS mitigation — filters the malicious flood while letting real users through.

Q11.13 Two parties share a secret key. An attacker can submit a specially crafted message and analyse the resulting ciphertext. Which technique compromises the key?

(a) Analysing patterns in intercepted messages (b) Comparing the crafted message's ciphertext with previously captured ciphertexts (c) Brute force (d) Matching message lengths

Answer: (b) — a **chosen-plaintext attack**. Related families worth knowing: known-plaintext, ciphertext-only, chosen-ciphertext, birthday, replay, side-channel.

Firewalls, segmentation, access

Q11.14 A firewall must filter on IP, port and protocol **and** track the state of active connections, without deploying multiple devices.

Answer: Stateful packet inspection firewall. The ladder: packet-filtering (stateless, L3/L4) → stateful → circuit-level gateway → application-layer/proxy (L7, deepest inspection).

Q11.15 A firewall must act as an **intermediary**, inspecting all incoming and outgoing traffic between the user's device and the internet.

Answer: Proxy firewall — it terminates and re-originates the connection, so it can inspect payloads.

Q11.16 In a web-filtering service the firewall holds the request, queries the web filter and allows or blocks accordingly. Which of the following is **not** a disadvantage?

Answer: "There is a trust relationship between the firewall and the web filter, so malicious activity on one affects the other." That describes normal architecture, not a functional disadvantage. The genuine disadvantages are the resource overhead of copying every request, and loss of connectivity if the filter fails.

Q11.17 What is the main security purpose of **network segmentation**?

Answer: Limiting the spread of attacks — it contains lateral movement. (Performance improvement is a side benefit, not the security purpose.)

Q11.18 Which cloud security measure requires two different types of identification before granting access?

Answer: Multi-factor authentication (MFA). The three factors: something you **know** (password), something you **have** (token, phone), something you **are** (biometric). **[HIGH-FREQ]**

Q11.19 What is the process of converting data into a secure format to prevent unauthorised access?

Answer: Encryption. Distinguish it from **authentication** (proving identity), **authorization** (granting permissions), **hashing** (one-way, irreversible) and **compression** (size, not security). **[HIGH-FREQ]**

VPNs

Five separate questions in the source papers all test the same three-way distinction. Learn the table once and you have answered all of them.

Scenario in the question	Answer
One employee, working from home, needs the internal network	Remote Access VPN
Office A and Office B, all users both ways, "as if on the same LAN"	Site-to-Site VPN
Encrypting and authenticating IP traffic between remote offices	IPSec VPN
On-premises datacentre + public cloud, hybrid model	Secure hybrid connectivity with VPN (or a dedicated link such as Direct Connect)

Q11.20 Employee working from home needs to retrieve files from internal servers. → **Remote Access VPN [HIGH-FREQ — three occurrences]**

Q11.21 All devices in Office A must reach resources in Office B over a shared network. → **Site-to-Site VPN [HIGH-FREQ]**

Q11.22 A network administrator with several remote offices wants secure remote access to the internal network. → **IPSec VPN**

Q11.23 An enterprise combining on-premises infrastructure with public cloud, needing data integrity and reliability. → **Secure hybrid connectivity with VPN [HIGH-FREQ — three occurrences]**

Q11.24 Which IPsec protocol provides both **confidentiality and integrity/authentication** for the packet payload and can run in tunnel mode?

(a) AH (b) IKE (c) ESP (d) GRE

Answer: (c) ESP. AH gives integrity and authentication but **no encryption**; IKE negotiates keys; GRE is a tunnelling protocol with no security of its own.

Q11.25 Which task **cannot** be outsourced when a company outsources antivirus management?

(a) Monitoring servers on premises (b) Preparing the internal antivirus **policy** (c) Rectifying virus infections (d) Updating definition files

Answer: (b) Preparing the internal policy. Operational work can be outsourced; **governance and policy ownership cannot**. This is the recurring IT-governance principle in Accenture papers.

Q11.26 In the shared responsibility model, a provider handles infrastructure services while the customer develops and deploys the application. How does this affect security management?

Answer: Both parties share security responsibility. The provider secures the cloud (hardware, hypervisor, physical facility); the customer secures what they put **in** the cloud (data, access control, application code, OS patching in IaaS).

Stage 12 — Cloud & DevOps

Service and deployment models

Q12.1 Which cloud service model provides hardware resources such as virtual machines and storage?

Answer: IaaS. The ladder: **IaaS** (you manage OS + apps) → **PaaS** (you manage apps only) → **SaaS** (you manage nothing; you just use it).

Q12.2 A growing business wants office productivity tools over the internet without IT management overhead.

Answer: SaaS — ready-to-use applications delivered over the internet.

Q12.3 Which cloud deployment model is typically most secure for highly sensitive data?

Answer: Private Cloud — dedicated to one organisation, maximum control. **[HIGH-FREQ]**

Q12.4 A company must keep sensitive data on-premises while using cloud resources for other workloads.

Answer: Hybrid Cloud. The four models: **Public, Private, Hybrid, Community** (shared by organisations with common concerns).

Storage, scaling and cost

Q12.5 Which cloud storage option scales dynamically without manual intervention?

(a) Block (b) File (c) Object (d) Table

Answer: (c) Object storage. Flat namespace, unstructured data, effectively unlimited scale (S3-style). **Block** storage = raw volumes attached to a VM (EBS-style); **file** storage = a shared filesystem (NFS-style). **[HIGH-FREQ]**

Q12.6 An e-commerce site hits sudden traffic spikes during a flash sale and must handle them without manual intervention. Which features?

Answer: Auto-scaling and load balancing. Auto-scaling adds and removes instances; load balancing spreads requests across them. **[HIGH-FREQ]**

Q12.7 How does **resource pooling** achieve elasticity in cloud storage across multi-vendor devices?

Answer: It aggregates disparate physical storage into one virtual pool, enabling dynamic allocation, automated failover and global-scale resilience. **[HIGH-FREQ]**

Q12.8 A startup is worried about unpredictable costs from variable cloud usage. Which feature helps predict spend?

(a) Auto-scaling (b) Reserved instances (c) Load balancing (d) CDN

Answer: (b) Reserved instances — a longer-term commitment at a discounted, predictable rate. Compare **on-demand** (flexible, expensive) and **spot** (cheapest, can be reclaimed).

Q12.9 A company migrating legacy data systems to the cloud needs consistent access to both old and new systems for real-time reporting. Which layer?

Answer: An abstraction layer that logically unifies disparate data sources.

Containers and virtualization

Q12.10 A team containerises an application to isolate it and its dependencies from the host **without** hardware-virtualization overhead. Which characteristic defines containers?

Answer: Containers share the host OS kernel — they are part of the same OS instance, unlike VMs which each need a guest OS under a hypervisor. **[HIGH-FREQ]**

The single most-asked comparison in these papers:

	Virtual Machine	Container
Isolation	Full — own guest OS	Process-level, shared kernel
Overhead	Heavy (GBs, slow boot)	Light (MBs, instant)
Can run a different OS?	Yes	No
Best for	Strong isolation, mixed OSes	Density, portability, fast deploys

Q12.11 In a cloud infrastructure running multiple service types, which service securely isolates a server's resources?

Answer: Virtual Machines.

DevOps practice

Q12.12 An e-commerce company updates its website weekly; after each update the new version is **automatically deployed to the live environment**. Which practice?

(a) Continuous Deployment (b) Continuous Integration (c) Continuous Testing (d) Monitoring

Answer: (a) Continuous Deployment. [HIGH-FREQ — three occurrences.] Keep the ladder straight:

- **CI** — merge and build code frequently, run automated tests on every commit
- **Continuous Delivery** — every build is *ready* to release, but a human presses the button
- **Continuous Deployment** — every passing build goes to production automatically
- **Continuous Testing** — automated tests run throughout the pipeline

The keyword "automatically deployed to the live environment" always means Continuous **Deployment**.

Q12.13 A team wants faster deployments and less downtime when updating a containerized app. Which practice?

Answer: Rolling updates with health checks — instances are replaced gradually and traffic only shifts to instances that pass their health check. Related strategies worth knowing: **blue-green** (two identical environments, flip the router) and **canary** (release to a small percentage first).

Stage 13 — MS Office / Common Applications

Source: Common Applications.pdf + Actual Paper MCQs.pdf. Easy marks — and the section most candidates skip and then lose.

Excel

Q13.1 Which Excel feature lets you create **custom calculations and summaries** within a PivotTable?

(a) Calculated Fields (b) Calculated Items (c) Slicers (d) Grouping

Answer: (a) Calculated Fields — a new formula-driven field built from existing fields. (A *Calculated Item* creates a new entry inside an existing field; a *Slicer* is a visual filter.) **[HIGH-FREQ]**

Q13.2 You must dynamically generate a list of employees from the South region (Excel 365). Which formula?

(a) `=IF(Region="South", B2:H9, "")` (b) `=FILTER(B2:H9, H2:H9="South")` (c) `=VLOOKUP("South", H2:H9, B2:H9)` (d) `=INDEX(B2:H9, MATCH("South", H2:H9, 0))`

Answer: (b) FILTER. It returns **all** matching rows as a dynamic array. VLOOKUP and INDEX/MATCH return a single match only.

Q13.3 `=INDEX(A2:A9, MATCH(MAX(C2:C9), C2:C9, 0))` over a product table where C is Unit Price. Result?

Answer: Laptop. Work it inside-out: `MAX(C2:C9) = 850` → `MATCH(850, C2:C9, 0) = position 1` → `INDEX(A2:A9, 1) = the first product`. **INDEX/MATCH inside-out evaluation is a guaranteed question — practise it.**

Q13.4 In Excel formulas, what does the notation `B8 : B10` indicate?

Answer: All cells from B8 to B10 (B8, B9, B10). The colon is the range operator; the comma is the union operator.

Word

Q13.5 Which paragraph formatting setting ensures a title and its paragraph stay on the same page?

(a) Wrap (b) Keep (c) Swap (d) Lock

Answer: (b) Keep — specifically **"Keep with next"**. Its sibling, "Keep lines together", stops a single paragraph being split across pages.

Q13.6 Which file format should you use to save a document **template** for repeated use?

(a) .docx (b) .pdf (c) .dotx (d) .txt

Answer: (c) .dotx. The pattern holds across Office: `.docx / .dotx`, `.xlsx / .xltx`, `.pptx / .potx` — the **t** means template.

Q13.7 Which shortcut inserts today's date in MS Word?

(a) Alt + Shift + D (b) Ctrl + Shift + T (c) Ctrl + Alt + D (d) Shift + F5

Answer: (a) Alt + Shift + D. Also know: **Alt + Shift + T** inserts the time; **Shift + F5** jumps to the last edit location.

Q13.8 Two statements about the Mini Translator in MS Word: (1) once activated it stays active until deactivated via the Translation menu; (2) when activated, the document is sent to Microsoft Translator as you type. Which are correct?

Answer: Both statements are correct.

Q13.9 Which MS Office application was used for creating and managing web pages?

Answer: FrontPage. (The source paper's question stem asks about auto-save intervals but marks FrontPage correct — a mismatched question and answer. See the errata.)

PowerPoint

Q13.10 You are building a training module and must force users to view each slide sequentially with narration, no free skipping. Which setting gives the strictest control?

(a) Hyperlinks between slides (b) Normal Slide Show mode (c) Browsed at a kiosk (full screen) (d) Presenter View

Answer: (c) Kiosk mode — disables manual navigation entirely. **[HIGH-FREQ]**

Q13.11 Jane wants a slide showing a progression over time with icons and text, visually dynamic, without animating each step manually. Which feature?

(a) Slide Zoom (b) SmartArt process layout (c) Morph transition across duplicated slides (d) Custom layout with animation triggers

Answer: (c) Morph transition. It automatically interpolates the movement of matched objects between two slides. **[HIGH-FREQ]**

Q13.12 A presenter wants to highlight bullet points one by one after explaining each, without animating every bullet separately. Best feature?

(a) Delay on each bullet (b) 'Appear' with a trigger (c) The **'By Paragraph'** animation option (d) Group and 'Wipe'

Answer: (c) 'By Paragraph'. One animation applied to the whole text box, revealed line by line via Effect Options. *(Note this is the opposite of Q13.11's answer — read whether the question says "progression across slides" (Morph) or "bullets within one text box" (By Paragraph).)*

Stage 14 — Game-Based / Logical Round

Source: *Accenture Game-Based.pdf*. This round tests working memory and planning under time pressure, not knowledge. You cannot revise it — you can only practise the strategy.

Game type 1 — Number ordering (≈25 games in the real test)

Compare three expressions and arrange in ascending order. Evaluate everything to a decimal before comparing. Watch for equal values.

#	Options	Correct order
1	12/5, 1.2, 2.3	1.2 < 2.3 < 2.4
2	3.5, 7/2, 3.25	3.25 < 3.5 = 3.5
3	5/4, 1.1+0.2, 1.6−0.1	1.25 < 1.3 < 1.5
4	2.9, 3−1/5, 14/5	2.8 = 2.8 < 2.9
5	√9, 2.8, 3.1−0.1	2.8 < 3.0 = 3.0
6	8/3, 2.5+0.1, 2.75	2.6 < 2.666 < 2.75
7	1.9×1.1, 2.05, 19/9	2.05 < 2.09 < 2.11
8	15/7, 2.1, 2.05+0.05	2.1 = 2.1 < 2.14
9	2³/5, 3.3−1, 7/2	1.6 < 2.3 < 3.5
10	√16/2, 2.1, 4/1.5	2.0 < 2.1 < 2.66

Strategy: convert every fraction to a decimal to 2–3 places first, *then* order. Most mistakes come from comparing a fraction directly against a decimal. Expect deliberate ties (games 2, 4, 5, 8) — the interface must accept an "equals" relation.

Game type 2 — Path / tile rotation

Rotate tiles 90° at a time to build a continuous path from S to E. Not all tiles need to be used;  is bidirectional (up/down).

Strategy: work **backwards from E** to find which direction must enter the exit, then forward from S. Count rotations before committing — each rotation is a move. Track your facing direction after every turn tile.

Game type 3 — Hidden path (invisible walls)

Start at S, collect the Key (K), reach the Exit (E). Walls (X) are invisible until you bump into them; a bump costs a move and reveals the cell. You cannot open E without the key.

Worked example (5×5, walls at (2,2), (2,4), (4,1), (4,3), (5,4)):

Start → Key: **D, D, R, R**. Key → Door: **R, D, R, D**. Full sequence: **D, D, R, R, R, D, R, D** — 8 moves after mapping.

Strategy: (1) Keep a mental map and mark every discovered wall — never re-test a known wall. (2) Prefer moves that reduce both the row and column distance to the target. (3) On a bump, do not retry a neighbouring cell of the same wall; change axis. (4) Remember the key comes first — routing straight to E wastes the run.

Game type 4 — Memory game

Same mechanic on a fogged grid where every unvisited cell shows . Each move reveals whether the new cell is safe or blocked.

Strategy: explore along **one edge** first — edges have fewer neighbours, so each discovery constrains more of the grid. Redraw the discovered map after every bump.

Game type 5 — Switch sequence

A 5×5 grid with coloured switches that must be activated in order (Red → Green → Blue → Yellow), invisible traps that send you back to Start, and a door that only opens once all four are active.

Strategy: plan the whole route before the first move, ordered by the required colour sequence rather than by proximity. Stepping on a wrong colour resets **all** switches, so a shorter path that risks the wrong order is a bad trade.

Stage 15 — Errata in the source PDFs

Read this before you memorise anything. These are places where the source papers are wrong or contradict each other. Getting them right matters more than another ten practice questions.

1. **Pseudo Code Q12 (here Q1.8)** — the header prints "Correct Answer: 26" but the PDF's own step-by-step trace concludes 13, and 13 is correct ($a > b \rightarrow 1 > 2$ is false). **Answer: 13.**
2. **Pseudo Code Q11 vs Q12** — both are labelled "Correct Answer: 26". Only Q11 is genuinely 26.
3. **Actual Paper MCQ Q16 (EIDE)** — the options are printed as "Enhanced Integrated Device Electronics A / B / C", which is garbled. The real option set is *Enhanced Interface Drive Electronics / Enhanced Integrated Drive Electronics / Enhanced Interface Driver Electronics / Enhanced Integrated Hard Driver*, and the answer is **Enhanced Integrated Drive Electronics (EIDE)**.
4. **OS property for varying hardware** — Actual Paper MCQ Q21 answers **Resource Allocation**; Common Applications Q9 answers **Device Management** for the same scenario. Device Management is the better answer when the stem emphasises differing *hardware specifications*.
5. **Routing manipulation attack** — Actual Paper MCQ Q14 answers **MITM**; MCQ Q29 answers **Routing Table Poisoning** for near-identical stems. Routing table poisoning is the specific mechanism; MITM is the category. **If both are options, choose Routing Table Poisoning.**
6. **WPA3 configuration** — Actual Paper MCQ Q19 says `Configure SUITEB192 = Y` is redundant; Networking Q19 says `CONFIG_SAE=y` is unnecessary. The **networking PDF is correct**: SAE belongs to WPA3-Personal, so it is the out-of-place line in a WPA3-Enterprise configuration.
7. **Actual Paper MCQ Q5** — the options list "MTP" (not a protocol here) and the answer text switches between "SSL/TLS" and "TLS". Either is accepted; TLS is the modern term.
8. **Common Applications Q15** — the stem asks about the MS Office feature that **auto-saves at intervals**, but the marked answer is **FrontPage**, which is a web-page editor. The question and answer do not match. The auto-save feature is **AutoRecover**; FrontPage is the answer to "which Office app created web pages".
9. **Actual Paper MCQ Q38** is missing entirely (the numbering jumps from Q37 to Q39).
10. **Networking Q37, Q38, Q39** print an empty "Correct answer :." line — the answers are, respectively, **stateful packet inspection firewall**, **Software as a Service**, and **Virtualization**, as their explanations state.
11. **Coding problem #13 (Sum of Distances)** — the formula is printed as $\text{sqrt}(x2-x1)^2 + (y2-y1)^2$. The correct formula is $\text{sqrt}((x2-x1)^2 + (y2-y1)^2)$.
12. **Actual Paper MCQ Q18** marks "Part of the same OS instance as the Hypervisor" as the container characteristic. Strictly, containers share the same OS instance as the **host**, not as a hypervisor — the phrasing is loose, but it is the intended answer.

End of question bank. Work through the companion *Topics & Learning Roadmap* document alongside this one — it maps every question here to the concept it tests and adds the surrounding material you are likely to be asked about but which these papers happen not to cover.